# Network Performance Monitoring and Tuning in Linux

There's nothing more annoying than a slow network connection. If you are a victim of poorly performing network devices, this article may help you to at least alleviate, if not solve, your problem.


The improvement in the performance of networking devices like switches, routers and connectors such as Ethernet cables has been huge in recent years. But our microprocessors lag behind when it comes to processing the data received and transferring it through them. Your organisation can spend money to buy high quality Ethernet cables or other networking devices to send terabytes or petabytes of data but may not experience performance improvements because, ultimately, the data has to be dealt with by the operating-system kernel. That is why, to get the most out of your fast Ethernet cable, as well as high bandwidth support and a high data transmission rate, it is important to tune your operating system. This article covers network monitoring and tuning methods—mostly parameters and settings available in the OS and network interface card (NIC)—that can be tweaked to improve the performance of your overall network.

## How your operating system deals with data

Data destined to a particular system is first received by the NIC and is stored in the ring buffer of the reception (RX) present in the NIC, which also has TX (for data transmission). Once the packet is accessible to the kernel, the device driver raises *softirq* (software interrupt), which makes the DMA (data memory access) of the system send that packet to the Linux kernel. The packet data in the Linux kernel gets stored in the *sk_buff* data structure to hold the packet up to MTU (maximum transfer unit). When all the packets are filled in the kernel buffer, they get sent to the upper processing layer – IP, TCP or UDP. The data then gets copied to the preferred data receiving process.

**Note:** Initially, a hard interrupt is raised by the device driver to send data to the kernel, but since this is an expensive task, it is replaced by a software interrupt. This is handled by the NAPI (new API), which makes the processing of incoming packets more efficient by putting the device driver in polling mode.

## Tools to monitor and diagnose your system's network

***ip:*** This tool shows/manipulates routing, devices, policy routing and tunnels (as mentioned in the man pages). It is used to set up and control the network.

***ip link or ip addr:*** This gives detailed information of all the interfaces. Other options and usage can be seen in *man ip*.